

	$\sigma_{1^+ a\beta}^{\#1}$	$\sigma_{1^+ a\beta}^{\#2}$	$\tau_{1^+ a\beta}^{\#1}$	$\sigma_{1^+ a}^{\#1}$	$\sigma_{1^+ a}^{\#2}$	$\tau_{1^+ a}^{\#1}$	$\tau_{1^+ a}^{\#2}$
$\sigma_{1^+}^{\#1} \uparrow^{\alpha\beta}$	$\frac{(2\beta_1-\beta_2)(1+k^2)}{\text{Det}(1^+)}$	$-\frac{4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}\text{Det}(1^+)}$	$-\frac{ik(4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2))}{2\sqrt{2}\text{Det}(1^+)}$	0	0	0	0
$\sigma_{1^+}^{\#2} \uparrow^{\alpha\beta}$	$-\frac{4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}\text{Det}(1^+)}$	$\frac{12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2)}{4(1+k^2)\text{Det}(1^+)}$	$\frac{ik(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2))}{4(1+k^2)\text{Det}(1^+)}$	0	0	0	0
$\tau_{1^+}^{\#1} \uparrow^{\alpha\beta}$	$\frac{ik(4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2))}{2\sqrt{2}\text{Det}(1^+)}$	$-\frac{ik(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2))}{4(1+k^2)\text{Det}(1^+)}$	$\frac{k^2(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2))}{4(1+k^2)\text{Det}(1^+)}$	0	0	0	0
$\sigma_{1^+}^{\#1} \uparrow^\alpha$	0	0	0	$\frac{(2\beta_1+\beta_2+\beta_3)(1+2k^2)}{2\text{Det}(1^+)}$	$\frac{2\beta_3+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}\text{Det}(1^+)}$	0	$\frac{ik(2\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{2\text{Det}(1^+)}$
$\sigma_{1^+}^{\#2} \uparrow^\alpha$	0	0	0	$\frac{2\beta_3+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}\text{Det}(1^+)}$	$\frac{4\beta_1+2\beta_2+4\beta_3+(\mathcal{M}_{\text{Pl}}^2)}{(4+8k^2)\text{Det}(1^+)}$	0	$\frac{i\sqrt{2}k(4\beta_1+2\beta_2+4\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{(4+8k^2)\text{Det}(1^+)}$
$\tau_{1^+}^{\#1} \uparrow^\alpha$	0	0	0	0	0	0	0
$\tau_{1^+}^{\#2} \uparrow^\alpha$	0	0	0	$-\frac{ik(2\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{2\text{Det}(1^+)}$	$-\frac{i\sqrt{2}k(4\beta_1+2\beta_2+4\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{(4+8k^2)\text{Det}(1^+)}$	0	$\frac{k^2(4\beta_1+2\beta_2+4\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{(2+4k^2)\text{Det}(1^+)}$

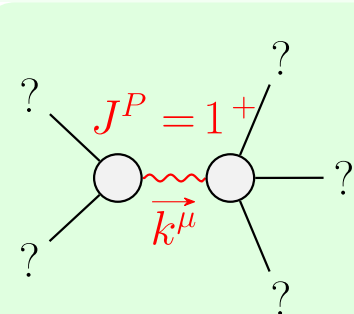
	$\mathcal{A}_{0^+}^{\#1}$	$f_{0^+}^{\#1}$	$f_{0^+}^{\#2}$	$\mathcal{A}_{0^+}^{\#1} \alpha\beta\chi$
$\mathcal{A}_{0^+}^{\#1} \uparrow$	$\frac{1}{2}(2\beta_1+\beta_2+3\beta_3+4(3\alpha_1+\alpha_3-\alpha_4+\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2))$	$-\frac{ik(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{\sqrt{2}}$	0	0
$f_{0^+}^{\#1} \uparrow$	$\frac{ik(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{\sqrt{2}}$	$(2\beta_1+\beta_2+3\beta_3)k^2$	0	0
$f_{0^+}^{\#2} \uparrow$	0	0	0	0
$\mathcal{A}_{0^+}^{\#1} \uparrow^{\alpha\beta\chi}$	0	0	0	$4\beta_1-4\beta_2+2(\alpha_2+3\alpha_4)k^2+\frac{(\mathcal{M}_{\text{Pl}}^2)}{2}$

$$\text{Det}(1^+) \rightarrow \frac{1}{8}(1+2k^2)(4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2))(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))$$

$$\text{Det}(0^+) \rightarrow 4\beta_1-4\beta_2+2(\alpha_2+3\alpha_4)k^2+\frac{(\mathcal{M}_{\text{Pl}}^2)}{2}$$

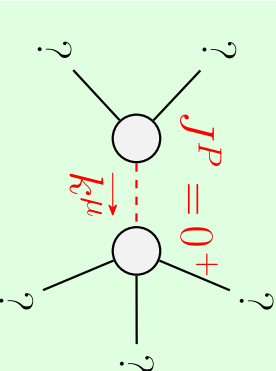
$$\text{Det}(2^+) \rightarrow \beta_1+\frac{\beta_2}{2}-\alpha_2k^2-\frac{(\mathcal{M}_{\text{Pl}}^2)}{4}$$

$$\text{Det}(2^+) \rightarrow \frac{1}{8}k^2(-4(3\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1+\beta_2)k^2+2(2\beta_1+\beta_2)(\mathcal{M}_{\text{Pl}}^2)-(\mathcal{M}_{\text{Pl}}^2)^2)$$

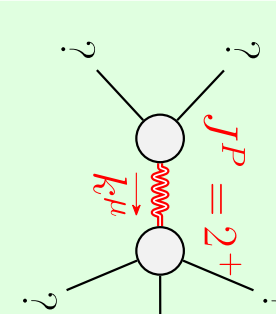


Massive particle	
Pole residue:	$(\alpha_2(48\beta_1^2-80\beta_1\beta_2+44\beta_2^2+8\beta_1(\mathcal{M}_{\text{Pl}}^2)-12\beta_2(\mathcal{M}_{\text{Pl}}^2)+(\mathcal{M}_{\text{Pl}}^2)^2)-\alpha_3(48\beta_1^2-80\beta_1\beta_2+44\beta_2^2+8\beta_1(\mathcal{M}_{\text{Pl}}^2)-12\beta_2(\mathcal{M}_{\text{Pl}}^2)+(\mathcal{M}_{\text{Pl}}^2)^2)+4\alpha_4(48\beta_1^2-80\beta_1\beta_2+44\beta_2^2+8\beta_1(\mathcal{M}_{\text{Pl}}^2)-12\beta_2(\mathcal{M}_{\text{Pl}}^2)+(\mathcal{M}_{\text{Pl}}^2)^2)-2(2\beta_1-\beta_2)(32\beta_1^2-16\beta_2^2+10\beta_2(\mathcal{M}_{\text{Pl}}^2)-(\mathcal{M}_{\text{Pl}}^2)^2-4\beta_1(4\beta_2+(\mathcal{M}_{\text{Pl}}^2))))/((\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1-\beta_2)(8\alpha_2\beta_1-8\alpha_3\beta_1+32\alpha_4\beta_1-32\alpha_6\beta_1-32\beta_1^2-4\alpha_2\beta_2+4\alpha_3\beta_2-16\alpha_4\beta_2+16\alpha_6\beta_2+16\beta_1\beta_2+16\beta_2^2+4\beta_1(\mathcal{M}_{\text{Pl}}^2)-10\beta_2(\mathcal{M}_{\text{Pl}}^2)+(\mathcal{M}_{\text{Pl}}^2)^2))>0$
Polarisations:	3
Square mass:	$\frac{-32\beta_1^2+16\beta_2^2-10\beta_2(\mathcal{M}_{\text{Pl}}^2)+(\mathcal{M}_{\text{Pl}}^2)^2+4\beta_1(4\beta_2+(\mathcal{M}_{\text{Pl}}^2))}{4(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1-\beta_2)}>0$
Spin:	1
Parity:	Even

Massive particle	
Pole residue:	$(-4\alpha_4\beta_1+4\alpha_6\beta_1-2\alpha_4\beta_2+2\alpha_6\beta_2-6\alpha_4\beta_3+6\alpha_6\beta_3-2\alpha_4(\mathcal{M}_{\text{Pl}}^2)+2\alpha_6(\mathcal{M}_{\text{Pl}}^2)+2\alpha_6(\mathcal{M}_{\text{Pl}}^2)+2\beta_1(\mathcal{M}_{\text{Pl}}^2)+\beta_2(\mathcal{M}_{\text{Pl}}^2)+3\beta_3(\mathcal{M}_{\text{Pl}}^2)+6\alpha_4(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))+2\alpha_3(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2)))/(2(3\alpha_1+\alpha_3-\alpha_4+\alpha_6)(2\beta_1+\beta_2+3\beta_3)(\mathcal{M}_{\text{Pl}}^2))>0$
Polarisations:	1
Square mass:	$\frac{(\mathcal{M}_{\text{Pl}}^2)(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{4(3\alpha_1+\alpha_3-\alpha_4+\alpha_6)(2\beta_1+\beta_2+3\beta_3)}>0$
Spin:	0
Parity:	Even

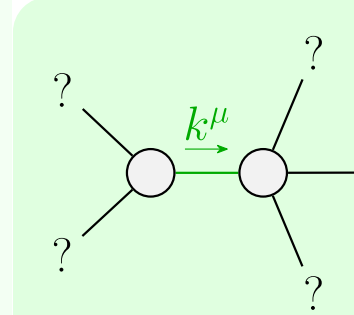


Massive particle	
Pole residue:	$(-3\alpha_2(4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2))+\alpha_3(4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2))-2(8\alpha_4\beta_1-8\alpha_6\beta_1+4\alpha_4\beta_2-4\alpha_6\beta_2-2\alpha_4(\mathcal{M}_{\text{Pl}}^2)+2\alpha_6(\mathcal{M}_{\text{Pl}}^2)+2\beta_1(\mathcal{M}_{\text{Pl}}^2)+\beta_2(\mathcal{M}_{\text{Pl}}^2)))/((3\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1+\beta_2)(\mathcal{M}_{\text{Pl}}^2))>0$
Polarisations:	5
Square mass:	$\frac{(4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2))(\mathcal{M}_{\text{Pl}}^2)}{4(3\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1+\beta_2)}>0$
Spin:	2
Parity:	Even



Massive particle	
Pole residue:	$-\frac{1}{2(\alpha_2+3\alpha_4)}>0$
Polarisations:	1
Square mass:	$\frac{8\beta_1-8\beta_2+(\mathcal{M}_{\text{Pl}}^2)}{4(\alpha_2+3\alpha_4)}>0$
Spin:	0
Parity:	Odd

Massive particle	
Pole residue:	$\frac{1}{\alpha_2}>0$
Polarisations:	5
Square mass:	$\frac{4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2)}{4\alpha_2}>0$
Spin:	2
Parity:	Odd



Quadratic pole	
Pole residue:	$\frac{1+8p^2}{(\mathcal{M}_{\text{Pl}}^2)}>0$
Polarisations:	2

Unitarity conditions
(Unitarity is demonstrably impossible)

	$\mathcal{A}_{1^+}^{\#1} \alpha\beta$	$\mathcal{A}_{1^+}^{\#2} \alpha\beta$	$f_{1^+}^{\#1} \alpha\beta$	$\mathcal{A}_{1^+}^{\#1} \alpha$	$\mathcal{A}_{1^+}^{\#2} \alpha$	$f_{1^+}^{\#1} \alpha$	$f_{1^+}^{\#2} \alpha$
$\mathcal{A}_{1^+}^{\#1} \uparrow^{\alpha\beta}$	$\frac{1}{4}(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2))$	$\frac{4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}}$	$\frac{ik(4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2))}{2\sqrt{2}}$	0	0	0	0
$\mathcal{A}_{1^+}^{\#2} \uparrow^{\alpha\beta}$	$\frac{4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}}$	$2\beta_1-\beta_2$	$i(2\beta_1-\beta_2)k$	0	0	0	0
$f_{1^+}^{\#1} \uparrow^{\alpha\beta}$	$-\frac{ik(4\beta_1-6\beta_2+(\mathcal{M}_{\text{Pl}}^2))}{2\sqrt{2}}$	$-i(2\beta_1-\beta_2)k$	$(2\beta_1-\beta_2)k^2$	0	0	0	0
$\mathcal{A}_{1^+}^{\#1} \uparrow^\alpha$	0	0	0	$\beta_1+\frac{\beta_2}{2}+\beta_3+\frac{(\mathcal{M}_{\text{Pl}}^2)}{4}$	$-\frac{2\beta_3+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}}$	0	$-\frac{1}{2}ik(2\beta_3+(\mathcal{M}_{\text{Pl}}^2))$
$\mathcal{A}_{1^+}^{\#2} \uparrow^\alpha$	0	0	0	$-\frac{2\beta_3+(\mathcal{M}_{\text{Pl}}^2)}{2\sqrt{2}}$	$\frac{1}{2}(2\beta_1+\beta_2+\beta_3)$	0	$\frac{i(2\beta_1+\beta_2+\beta_3)k}{\sqrt{2}}$
$f_{1^+}^{\#1} \uparrow^\alpha$	0	0	0	0	0	0	0
$f_{1^+}^{\#2} \uparrow^\alpha$	0	0	0	$\frac{1}{2}ik(2\beta_3+(\mathcal{M}_{\text{Pl}}^2))$	$-\frac{i(2\beta_1+\beta_2+\beta_3)k}{\sqrt{2}}$	0	$(2\beta_1+\beta_2+\beta_3)k^2$

	$\sigma_{2^+}^{\#1} \alpha\beta$	$\tau_{2^+}^{\#1} \alpha\beta$	$\sigma_{2^+}^{\#1} \alpha\beta\chi$
$\sigma_{2^+}^{\#1} \uparrow^{\alpha\beta}$	$\frac{(2\beta_1+\beta_2)k^2}{\text{Det}(2^+)}$	$\frac{ik(4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2))}{2\sqrt{2}\text{Det}(2^+)}$	0
$\tau_{2^+}^{\#1} \uparrow^{\alpha\beta}$	$-\frac{ik(4\beta_1+2\beta_2-(\mathcal{M}_{\text{Pl}}^2))}{2\sqrt{2}\text{Det}(2^+)}$	$\frac{4\beta_1+2\beta_2+2(-3\alpha_2+\alpha_3-4\alpha_4+4\alpha_6)k^2-(\mathcal{M}_{\text{Pl}}^2)}{4\text{Det}(2^+)}$	0
$\sigma_{2^+}^{\#1} \uparrow^{\alpha\beta\chi}$	0	0	$\frac{1}{\text{Det}(2^+)}$

$$\text{Det}(0^+) \rightarrow \frac{1}{2}k^2(4(3\alpha_1+\alpha_3-\alpha_4+\alpha_6)(2\beta_1+\beta_2+3\beta_3)k^2-(2\beta_1+\beta_2+3\beta_3)(\mathcal{M}_{\text{Pl}}^2)-(\mathcal{M}_{\text{Pl}}^2)^2)$$

$$\text{Det}(1^+) \rightarrow \frac{1}{8}(1+k^2)(16(\beta_1-\beta_2)(2\beta_1+\beta_2)+4(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1-\beta_2)k^2-4\beta_1(\mathcal{M}_{\text{Pl}}^2)+10\beta_2(\mathcal{M}_{\text{Pl}}^2)-(\mathcal{M}_{\text{Pl}}^2)^2)$$

	$\sigma_{0^+}^{\#1}$	$\tau_{0^+}^{\#1}$	$\tau_{0^+}^{\#2}$	$\sigma_{0^+}^{\#1} \alpha\beta\chi$
$\sigma_{0^+}^{\#1} \uparrow$	$\frac{(2\beta_1+\beta_2+3\beta_3)k^2}{\text{Det}(0^+)}$	$\frac{ik(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{\sqrt{2}\text{Det}(0^+)}$	0	0
$\tau_{0^+}^{\#1} \uparrow$	$-\frac{ik(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{\text{Pl}}^2))}{\sqrt{2}\text{Det}(0^+)}$	$\frac{2\beta_1+\beta_2+3\beta_3+4(3\alpha_1+\alpha_3-\alpha_4+\alpha_6)k^2+(\mathcal{M}_{\text{Pl}}^2)}{2\text{Det}(0^+)}$	0	0
$\tau_{0^+}^{\#2} \uparrow$	0	0	0	0
$\sigma_{0^+}^{\#1} \uparrow^{\alpha\beta\chi}$	0	0	0	$\frac{1}{\text{Det}(0^+)}$

Source constraints	#
SO(3) irreps	1
$\tau_{0^+}^{\#2} = 0$	1
$\tau_{1^+}^{\#2} \alpha + 2ik\sigma_{1^+}^{\#2} \alpha = 0$	3
$\tau_{1^+}^{\#1} \alpha = 0$	3
$\tau_{1^+}^{\#1} \alpha\beta + ik\sigma_{1^+}^{\#2} \alpha\beta = 0$	3
Total #:	10

Lagrangian density	
$2\beta_1\mathcal{A}_{1\theta\kappa}\mathcal{A}^{\theta\kappa}-\beta_2\mathcal{A}_{1\theta\kappa}\mathcal{A}^{\theta\kappa}-2\beta_1\mathcal{A}_{1\kappa\theta}\mathcal{A}^{\theta\kappa}+3\beta_2\mathcal{A}_{1\kappa\theta}\mathcal{A}^{\theta\kappa}-\frac{1}{2}(\mathcal{M}_{\text{Pl}}^2)\mathcal{A}_{1\kappa\theta}\mathcal{A}^{\theta\kappa}-\beta_3\mathcal{A}^{\theta}_{\theta}\mathcal{A}^{\theta\kappa}_{\theta\kappa}-\frac{1}{2}(\mathcal{M}_{\text{Pl}}^2)\mathcal{A}^{\theta}_{\theta}\mathcal{A}^{\theta\kappa}_{\theta\kappa}-\alpha_2\partial_{\lambda}\mathcal{A}^{\kappa\lambda}_{\kappa}\partial^{\lambda}\mathcal{A}_{\lambda}^{\zeta}+\beta_3\mathcal{A}^{\kappa}_{\kappa}\partial_{\theta}f^{\theta}-\beta_3\mathcal{A}^{\kappa\lambda}_{\kappa}\partial_{\theta}f_{\lambda}^{\theta}-(\mathcal{M}_{\text{Pl}}^2)f^{\theta}\partial_{\theta}\mathcal{A}_{\theta}\mathcal{A}^{\kappa}_{\kappa}+(\mathcal{M}_{\text{Pl}}^2)\partial_{\theta}\mathcal{A}^{\theta}_{\theta}+2\alpha_2\partial^{\alpha}\mathcal{A}^{\theta\zeta}_{\kappa}\partial_{\theta}\mathcal{A}_{\zeta\alpha}^{\kappa}+4\alpha_4\partial^{\alpha}\mathcal{A}^{\theta\zeta}_{\kappa}\partial_{\theta}\mathcal{A}_{\zeta\alpha}^{\kappa}-\beta_3\mathcal{A}_{\theta\kappa}^{\kappa}\partial^{\theta}f'_{\theta}+\beta_3\mathcal{A}_{\theta\kappa}^{\kappa}\partial^{\theta}f'_{\theta}+\beta_3\partial_{\theta}f_{\kappa}^{\kappa}\partial^{\theta}f'_{\theta}+(\mathcal{M}_{\text{Pl}}^2)f^{\theta}\partial_{\theta}\mathcal{A}_{\theta}\mathcal{A}^{\kappa}_{\kappa}-(\mathcal{M}_{\text{Pl}}^2)f'_{\theta}\partial_{\kappa}\mathcal{A}^{\theta\kappa}_{\theta}-\alpha_3\partial_{\theta}\mathcal{A}_{\lambda}^{\zeta}\partial_{\kappa}\mathcal{A}^{\theta\kappa\lambda}-2\alpha_2\partial_{\theta}\mathcal{A}_{\zeta\alpha}^{\kappa}\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}-(\mathcal{M}_{\text{Pl}}^2)f'_{\theta}\partial_{\kappa}\mathcal{A}^{\theta\kappa}_{\theta}-\alpha_3\partial_{\theta}\mathcal{A}_{\lambda}^{\zeta}\partial_{\kappa}\mathcal{A}^{\theta\kappa\lambda}-2\alpha_2\partial_{\theta}\mathcal{A}_{\zeta\alpha}^{\kappa}\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}+4\alpha_4\partial_{\theta}\mathcal{A}_{\zeta\alpha}^{\kappa}\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}+2\alpha_2\partial_{\theta}\mathcal{A}_{\lambda}^{\zeta}\partial_{\kappa}\mathcal{A}^{\theta\kappa\lambda}-2\alpha_4\partial_{\theta}\mathcal{A}_{\zeta\alpha}^{\kappa}\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}+2\beta_1\mathcal{A}_{1\theta\kappa}\partial^{\kappa}f^{\theta}-\beta_2\mathcal{A}_{1\theta\kappa}\partial^{\kappa}f^{\theta}+2\beta_2\mathcal{A}_{1\kappa}^{\lambda}\partial^{\kappa}f'_{\lambda}+2\beta_1\mathcal{A}_{\theta\kappa}^{\lambda}\partial^{\kappa}f_{\lambda}^{\theta}-\beta_2\mathcal{A}_{\theta\kappa}^{\lambda}\partial^{\kappa}f_{\lambda}^{\theta}-2\beta_1\mathcal{A}_{\theta\kappa}^{\lambda}\partial^{\kappa}f_{\lambda}^{\theta}+\beta_2\mathcal{A}_{\theta\kappa}^{\lambda}\partial^{\kappa}f_{\lambda}^{\theta}+\beta_2\mathcal{A}_{\theta\kappa}^{\lambda}\partial^{\kappa}f_{\lambda}^{\theta}+2\beta_1\partial_{\kappa}f_{\lambda}^{\theta}\partial^{\kappa}f_{\lambda}^{\theta}+2\beta_1\mathcal{A}_{\theta\kappa}^{\zeta}\partial^{\kappa}f_{\zeta}^{\theta}-\beta_2\mathcal{A}_{\theta\kappa}^{\zeta}\partial^{\kappa}f_{\zeta}^{\theta}+2\alpha_2\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}\partial^{\kappa}\mathcal{A}_{\zeta\alpha\theta}+4\alpha_4\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}\partial^{\kappa}\mathcal{A}_{\zeta\alpha\theta}+2\alpha_4\partial_{\kappa}\mathcal{A}^{\theta\zeta\alpha}\partial^{\kappa}\mathcal{A}_{\zeta\alpha\theta}-2\beta_3\partial^{\theta}f'_{\theta}\partial_{\lambda}f_{\theta}^{\lambda}+\beta_3\partial_{\theta}f_{\kappa}^{\theta}\partial_{\lambda}f^{\kappa\lambda}+4\alpha_1\partial_{\theta}\mathcal{A}^{\theta}_{\theta}\partial_{\lambda}\mathcal{A}^{\kappa\lambda}_{\kappa}+4\alpha_6\partial^{\alpha}\mathcal{A}_{\lambda}^{\lambda\zeta}\partial_{\lambda}\mathcal{A}_{\zeta\alpha}^{\alpha}-2\alpha_2\partial^{\alpha}\mathcal{A}_{\lambda}^{\zeta\lambda}\partial_{\lambda}\mathcal{A}_{\zeta\alpha}^{\alpha}-4\alpha_4\partial^{\alpha}\mathcal{A}_{\lambda}^{\zeta\lambda}\partial_{\lambda}\mathcal{A}_{\zeta\alpha}^{\alpha}+\alpha_3\partial_{\theta}\mathcal{A}_{\lambda}^{\zeta}\partial^{\lambda}\mathcal{A}^{\theta\kappa}_{\kappa}+\alpha_3\partial_{\kappa}\mathcal{A}^{\theta\kappa\lambda}\partial_{\zeta}\mathcal{A}_{\lambda}^{\zeta}_{\theta}-\alpha_2\partial_{\kappa}\mathcal{A}^{\kappa\lambda\theta}\partial_{\zeta}\mathcal{A}_{\lambda}^{\zeta}_{\theta}-\alpha_3\partial^{\lambda}\mathcal{A}^{\theta\kappa}_{\kappa}\partial_{\zeta}\mathcal{A}_{\lambda}^{\zeta}_{\theta}-\beta_2\partial^{\kappa}f_{\zeta}^{\theta}\partial^{\zeta}f_{\theta\kappa}-2\beta_1\partial^{\kappa}f_{\zeta}^{\theta}\partial^{\zeta}f_{\kappa\theta}+\beta_2\partial^{\kappa}f_{\lambda\zeta}\partial^{\zeta}f_{\kappa}^{\lambda}-\beta_2\partial^{\kappa}f_{\zeta\lambda}\partial^{\zeta}f_{\kappa}^{\lambda}$	
Added source term(s):	$f^{\alpha\beta}\tau_{\alpha\beta}+\mathcal{A}^{\alpha\beta\chi}\sigma_{\alpha\beta\chi}$